Project 70: AI Podcast Summarizer

Goal: Build a pipeline that:

1. **Transcribes** a podcast using Whisper
2. **Chunks and embeds** the transcript
3. **Summarizes** key sections using a local or cloud LLM
4. Outputs a clean **summary or episode notes**

Components

Step

Tool/Model

Audio to Text

`OpenAI Whisper` (`base` or `large`)

Chunking

Text split by tokens or timestamps

Summarizer Agent

`Transformers` (`T5`, `Falcon`, `GPT`)

Optional Memory

`ChromaDB`, `FAISS`, or LangChain-style cache

Step-by-Step Implementation

Step 1: Install Dependencies

```
pip install openai-whisper nltk transformers
```

Step 2: Transcribe the Podcast

```python
import whisper

# Load Whisper and transcribe
model = whisper.load_model("base")  # Or "medium"/"large" for better accuracy
result = model.transcribe("podcast_episode.mp3")

transcript = result["text"]
print("🎧 Transcription complete.")
```

Step 3: Chunk the Transcript for Long-Form Input

```python
import nltk
nltk.download('punkt')
from nltk.tokenize import sent_tokenize

def chunk_text(text, max_words=200):
    sentences = sent_tokenize(text)
    chunks = []
    current = ""
    for sent in sentences:
        if len(current.split()) + len(sent.split()) < max_words:
            current += " " + sent
        else:
            chunks.append(current.strip())
            current = sent
    chunks.append(current.strip())
```

```
    return chunks

chunks = chunk_text(transcript)
```

Step 4: Summarize Each Chunk Using an LLM

```
from transformers import pipeline

summarizer = pipeline("summarization", model="sshleifer/distilbart-cnn-12-6")

summaries = []
for i, chunk in enumerate(chunks):
    print(f"🔍 Summarizing chunk {i+1}/{len(chunks)}...")
    summary = summarizer(chunk, max_length=100, min_length=30, do_sample=False
    summaries.append(summary)
```

Step 5: Combine Summaries into a Final Report

```
final_summary = "\n\n".join(summaries)
print("\n📝 Final Podcast Summary:\n")
print(final_summary)
```

Key Concepts

- **Chunking** avoids hitting token limits in LLMs
- **Summarization models** condense each section
- Combines ASR (Whisper) + NLP summarization = complete pipeline

Project Summary

- **Input**: Long audio podcast (`.mp3` , `.wav`)
- **Process**: Transcribe → Chunk → Summarize
- **Output**: Concise summary or newsletter content
- **Use Case**: Podcast producers, note-taking, show notes, accessibility